

Section 7.0 – Geometry Intro

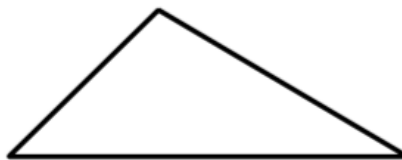
MPM1D

Part 1: Classifying Triangles

Classify Using Side Lengths:

Scalene Triangle

- no equal sides or angles



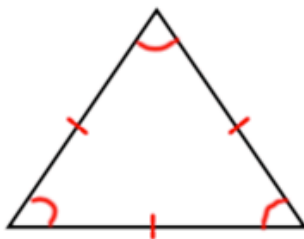
Isosceles Triangle

- 2 equal sides
- 2 equal angles



Equilateral Triangle

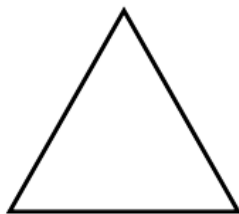
- 3 equal sides
- 3 equal angles



Classify Using Angle Measures:

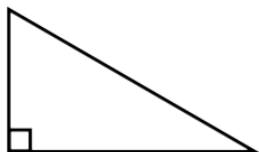
Acute Triangle

- 3 acute angles
(less than 90 degrees)



Right Triangle

- one right angle
(90 degrees)



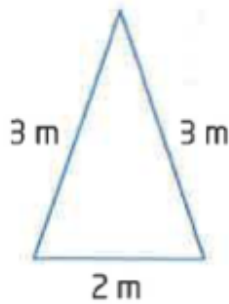
Obtuse Triangle

- one obtuse angle
(between 90 and 180 degrees)

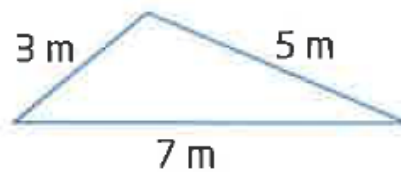


Example 1: Classify Each Triangle Using its Side Lengths

a)

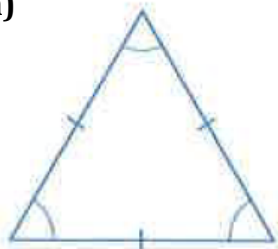


b)



Example 2: Classify Each Triangle in Two ways using its angle measures

a)



b)



Part 2: Classifying Polygons

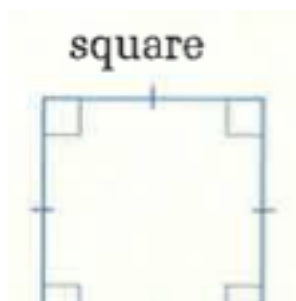
A **polygon** is a closed figure formed by three or more line segments.

A **regular polygon** has all sides equal and all angles equal.

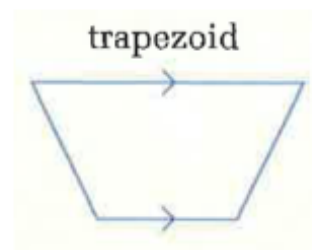
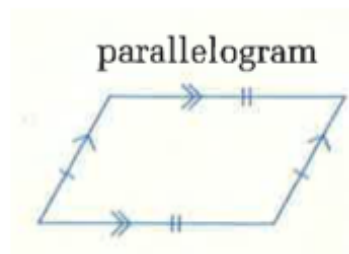
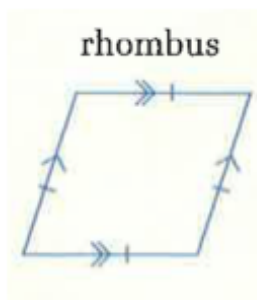
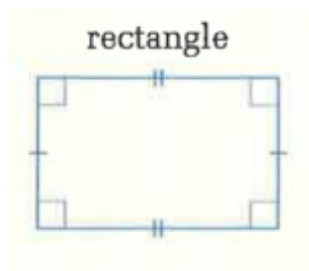
Some **quadrilaterals** have special names.

A **regular** quadrilateral is a square.

Number of Sides	Name
3	triangle
4	quadrilateral
5	pentagon
6	hexagon

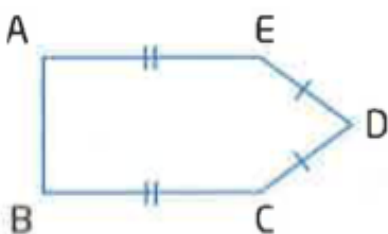


An **irregular** quadrilateral may be a **rectangle**, **rhombus**, **parallelogram**, or **trapezoid**

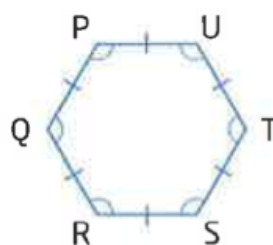


Example 3: Classify each polygon according to its number of sides and whether it is regular or irregular.

a)

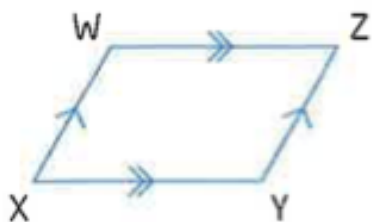


b)

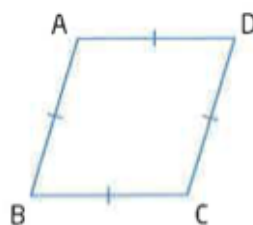


Example 4: Classify each quadrilateral

a)



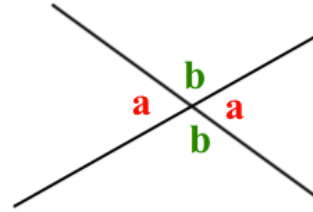
b)



Part 3: Angle Properties

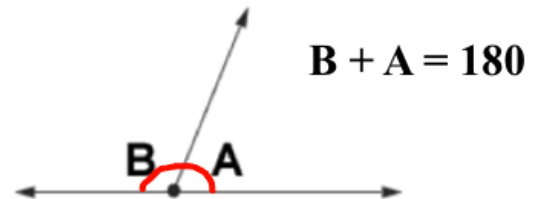
Opposite Angles:

- When 2 angles intersect, the opposite angles are equal.



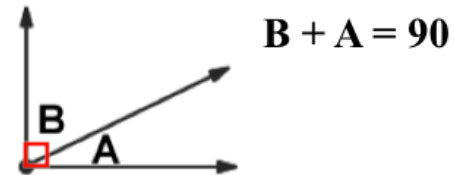
Supplementary Angles:

- angles that add to 180 degrees
- angles on a straight line are supplementary



Complementary Angles:

- angles that add to 90 degrees

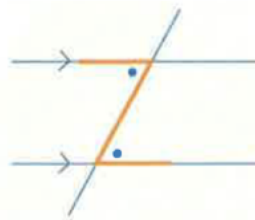


Part 4: Parallel Line Theorems

When a transversal crosses parallel lines, many pairs of angles are related...

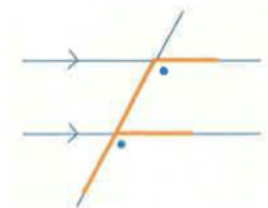
Alternate Interior Angles are equal

- Z pattern



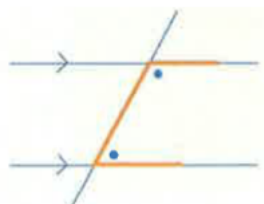
Corresponding Angles are equal

- F pattern



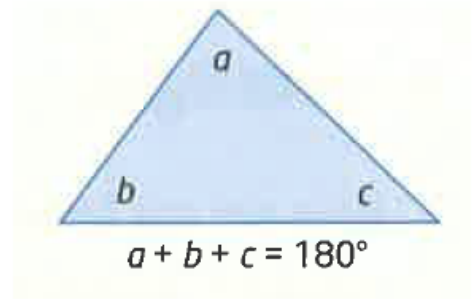
Co-Interior Angles add to 180

- C pattern

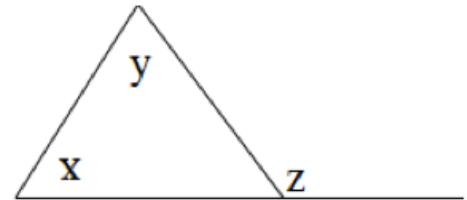


Part 6: Triangle Theorems

The sum of the **interior angles** of a triangle is **180** degrees.

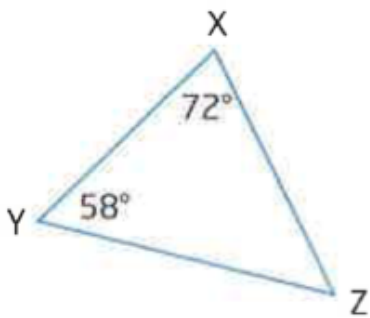


The **exterior angle** is equal to the sum of the 2 opposite interior angles.

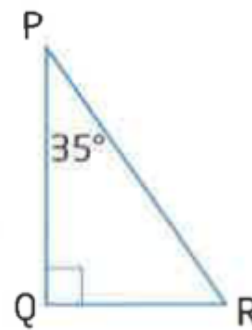


Example 5: Find the measure of the third angle in each triangle...

a)

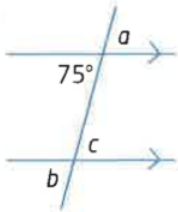


b)



Example 6: Find the measure of the angles a, b, and c. Give reasons for your answers...

a)



b)

